

Machine Learning for Bioengineering: Introduction to the Course

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Office hours:

- Before or after the lesson
- On Wednesday from 10:00 a.m. to 12:00 p.m. during the teaching period meetings can be arranged at my office in Bologna. However, students must request an appointment in advance, as I also have students at the Cesena Campus.
- By appointment: please email me to schedule a TEAMS meeting
- More than a week before examinations

Machine Learning for Bioengineering: organization of the teaching

- **60 hours** (lectures + laboratory sessions)
- Module 1 - First **30 hours** (prof. Diciotti)
- Module 2 - Last **30 hours** (prof. Furini)

ORARIO I ANNO LM

II periodo (dal 16/02/2026 al 05/06/2026)

Laurea Magistrale in Biomedical Engineering Curriculum Biomedical Engineering for Neuroscience					
Ore	Lunedì	Martedì	Mercoledì	Giovedì	Venerdì
9:00-10:00	BioSysMod (2.1)	BioInst (3.7)		BioInst (2.1)	
10:00-11:00	BioSysMod (2.1)	BioInst (3.7)		BioInst (2.1)	
11:00-12:00	BioSysMod (2.1)	BioInst (3.7)	BioSysMod (2.9)	MLBio (2.2)	
12:00-13:00			BioSysMod (2.9)	MLBio (2.2)	
13:00-14:00	BioSignal (2.6)	BioSignal (3.7/3.1)*			
14:00-15:00	BioSignal (2.6)	BioSignal (3.7/3.1)	MLBio (3.3)	BioSysMod (4.2)	
15:00-16:00	BioSignal (2.6)		MLBio (3.3)	BioSysMod (4.2)	
16:00-17:00			MLBio (3.3)		
17:00-18:00					

*laboratorio informatico dalla terza settimana

- Course materials, announcements, and additional resources available on virtuale.unibo.it

Modules 1 and 2 of safety training required

Important remark

In view of the type of activities and teaching methods adopted, the attendance of this training activity requires the prior participation of all students in Modules 1 and 2 of safety training in the workplace, in e-learning mode.

Course feedback questionnaire

- Your feedback is valuable throughout the course, not just at the end!
- Help improve the learning experience by sharing your thoughts and suggestions

[illegible]

One written examination for Machine Learning for Bioengineering (modules 1 and 2 together)

Exam dates and locations

- **08/06/2026, 14.00** — Lab 4.2
- **22/06/2026, 9:30** — Lab 2.2
- **23/07/2026, 9:30** — Lab 2.2
- **08/09/2026, 14:00** — Lab 4.2

Registration and attendance rules

- Registration opens the day after the previous exam session (except for the first exam).
- If you are unable to attend, please unsubscribe from the list.

One written examination for Machine Learning for Bioengineering (modules 1 and 2 together)

- Module 1 - Prof. Diciotti
 - 1 Theoretical question
 - 1 Exercise on the PC
- Module 2 - Prof. Furini
 - 1 Theoretical question
 - 1 Exercise on the PC
- Overall grade is the average of the two grades
- If the overall grade is insufficient, the student has to repeat the examination for both modules
- It is allowed to reject the overall grade only once

- **Biomedical Signal Processing (6 CFU)** - Prof. Davide Borra
- **Machine Learning for Bioengineering (6 CFU)** - Prof. Stefano Diciotti (module 1) and Prof. Simone Furini (module 2)

Final grade calculation method - Biomedical Signal Processing and Machine Learning (I.C.)

- Separate exam dates for the two courses (Biomedical Signal Processing & Machine Learning for Bioengineering).
- The grade for a single course is valid for 6 months.
- The final grade for Biomedical Signal Processing and Machine Learning (Integrated Course) is the average of the two grades rounded up, except when aiming for a final grade of 30 or 30 with honors. In these cases, students must achieve a grade of 30 in both modules or 30 with honors in both modules.
- **Examples:**
 - 27 and 28 $\rightarrow$ average = 27.5 $\rightarrow$ final grade = 28
 - 29 and 30 $\rightarrow$ average = 29.5 $\rightarrow$ final grade = 29
 - 30 and 30 $\rightarrow$ average = 30 $\rightarrow$ final grade = 30

At the end of the course, the student

- knows the main theoretical and practical tools for biomedical data analysis through machine learning algorithms
- understands the fundamental programming principles of the Python language and is able to apply them primarily to data management and analysis
- understand the role, purpose and features of Python libraries for numerical computation, data representation, and machine learning, and their interconnectivity
- has the ability to deepen further innovative topics by evaluating their pros and cons

- The exam is designed to assess the achievement of the main learning objectives of the course.
- In particular, it evaluates:
 - theoretical and practical knowledge of Machine Learning methods for Bioengineering.
 - the ability to apply methods to problem-solving scenarios.
 - analytical and critical reasoning skills.
 - clarity of exposition and appropriate use of scientific language and terminology.

Why machine learning tailored to medical data?

- Sometimes, machine learning systems for non medical data can be adapted for medical data, but sometimes they must be tailored to medical data (e.g., for data typical of the biomedical field, e.g., 3D medical imaging, genetic data, etc.)
- Machine learning techniques have to handle the shortage of medical data
- Medical data are usually protected by restricted data access, and distributed machine learning algorithms that do not require data pooling may be needed

- **Basics of AI theory** Understanding the fundamental principles behind AI and its development and application in Bioengineering and Medicine
- **Big picture understanding** Gaining a comprehensive perspective on how AI projects in Bioengineering and Medicine are structured, from data acquisition to model deployment, with an emphasis on interpretability and clinical impact
- **Hands-on implementation** Developing practical skills by working with real-world datasets and AI tools to build applications with biomedical relevance

Course contents (module 1)

1. Theoretical foundations of machine learning

- Definition and core concepts of Machine Learning.
- Taxonomy of Machine Learning algorithms (supervised, unsupervised, semi-supervised learning).
- Model capacity and generalization.
- Overfitting and regularization techniques.

2. Model optimization and validation strategies

- Hyperparameter tuning.
- Hold-out validation.
- Cross-validation.
- Nested hold-out and nested cross-validation.
- Leave-one-out and stratified sampling strategies.

3. Computational tools and reproducible workflows

- Introduction to Python programming for Machine Learning.
- Version control and collaborative development with GitHub.
- Cloud-based implementation and reproducibility using Google Colab.

4. Application to Bioengineering: Neuroimaging examples

- Age prediction task using brain Magnetic Resonance Imaging (MRI) data: development of an end-to-end Machine Learning pipeline.

Students' presentation

- Open-source Magnetic Resonance Imaging (MRI) TableTop
- Swarm learning
- Machine and Deep Learning in neuroimaging
- Synthetic MRI data generation
- AI for lung cancer screening
- Self-supervised learning for biomedical data
- Integrating biological and artificial learning

- A powerful and versatile method, with the ability to safely and noninvasively measure a wide range of properties within the living brain
- Capable of capturing images that provide information about gross anatomy, neuronal activity, connectivity, pathologies, and more

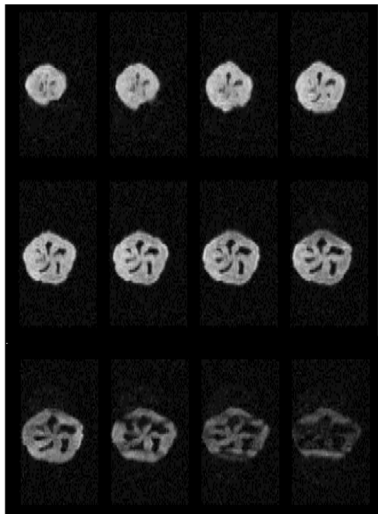


MRI TableTop

- 0.28T @isocenter permanent magnet
- RF single channel coil
- TX/RX switch
- RF power amplifier
- X, Y, Z Gradient coils
- Z^2 shim coil with active eddy current suppression



MRI TableTop - 3D imaging



MRI TableTop @Cesena Campus

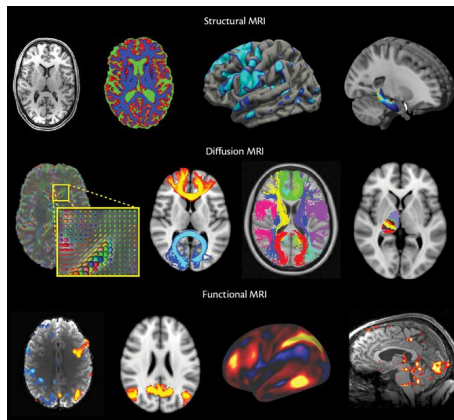


Play video (external viewer)

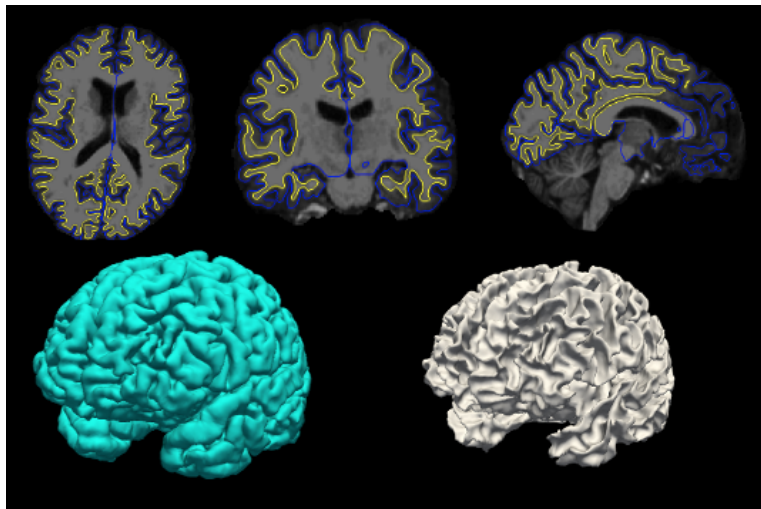
Play video (external viewer)

Main MRI modalities and analysis techniques

- Different type of images capture different information about the brain and are also called modalities. Three modalities are by far the most commonly used in neuroimaging research: structural, diffusion, and functional imaging
- Structural imaging for gross anatomical structures
- Diffusion imaging provides information about microstructure and anatomical connectivity
- Functional imaging or functional MRI (fMRI) provides information about the activity of the neurons in the brain



Machine and deep learning in Neuroimaging



Machine and deep learning in Neuroimaging: features

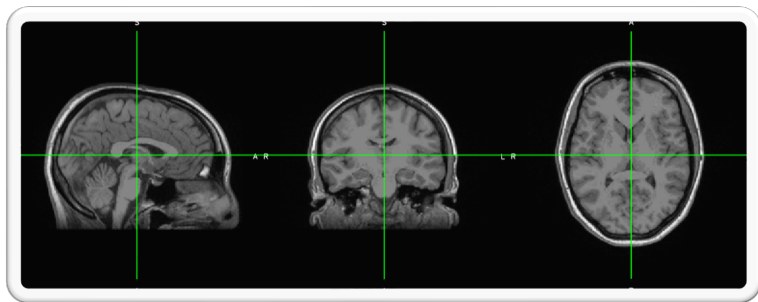


Figure: Sagittal, coronal and axial views

Machine and deep learning in Neuroimaging: features



Figure: Cortical parcellation

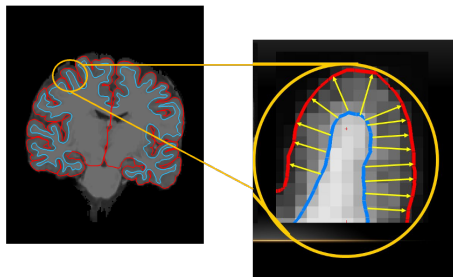


Figure: Cortical thickness

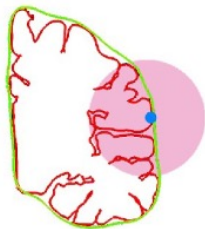
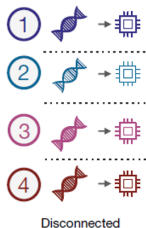


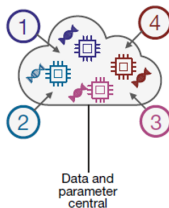
Figure: Local gyrification index

Swarm learning

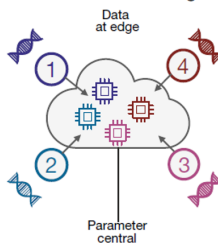
a Local learning



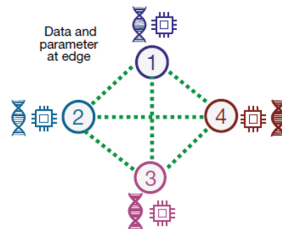
b Central learning



c Federated learning



d Swarm Learning



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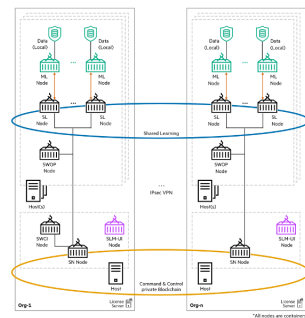
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Swarm Learning for decentralized and confidential clinical machine learning

[Stefanie Wornat-Herresthal](#), [Hartmut Schultze](#), [Krishnaprasad Lingadahalli Shastry](#), [Sathyanarayanan Manamohan](#), [Saikat Mukherjee](#), [Vishesh Garg](#), [Ravi Sarveswara](#), [Kristian Händler](#), [Peter Pickkers](#), [N. Ahmad Aziz](#), [Sofia Ktena](#), [Florian Tran](#), [Michael Bitzer](#), [Stephan Ossowski](#), [Nicolas Casadei](#), [Christian Herr](#), [Daniel Petersheim](#), [Uta Behrends](#), [Fabian Kern](#), [Tobias Fehlmann](#), [Philipp Schommers](#), [Clara Lehmann](#), [Max Augustin](#), [Jan Rybníček](#), [COVID-19 Aachen Study \(COVAS\)](#), [Deutsche COVID-19 Omics Initiative \(DeCOI\)](#), ... [Joachim L. Schultze](#) ✉ [+ Show authors](#)

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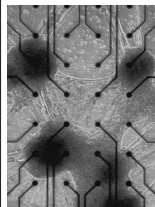
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The Future is Biological – Introducing the CL1

The CL1 by Cortical Labs is the world's first commercially available biological computer, a self-contained system that integrates a neural interface, perfusion life support, and on-device computation.

A powerful Python API allows users to design experiments and "program" the neural network in real time.



Play video (external viewer)

Suggested reading and learning resources

- Notes and scientific articles provided by the Professor

Machine Learning Foundations

- Goodfellow I., Bengio Y., Courville A. (2016). *Deep Learning*. MIT Press.
Ch. 1 (Introduction); Ch. 5 (Machine Learning Basics), Sections 5.1–5.3
<https://www.deeplearningbook.org/>

Python for Beginners

- Google Python Introduction:
<https://developers.google.com/edu/python/introduction>
- LearnPython.org: <https://www.learnpython.org/>
- Lutz M. (2013). *Learning Python*. O'Reilly.
(Available at Biblioteca Centrale Campus Cesena)
- W3Schools Python Tutorials (with exercises):
<https://www.w3schools.com/python/default.asp>

Official Python Documentation

- Python: <https://docs.python.org/3/>
- NumPy: <https://numpy.org/doc/>
- Pandas: <https://pandas.pydata.org/docs/>
- Matplotlib: <https://matplotlib.org/>
- Scikit-learn: <https://scikit-learn.org/stable/>

Python for Machine Learning (Optional)

- Géron A. (2019). *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*. O'Reilly.
(Available at Biblioteca Centrale Campus Cesena)